AI-ASSISTED CODE ENFORCEMENT

For Residential Violations

Client: Ryan Chelton, Director of Development Services, Town of Riverdale Park
Course: University of Maryland | INST 623 | Spring 2026
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1. Roles and Responsibilities

Effective project management requires not only technical competence but also clear accountability structures that align with the scope and demands of the work. This section outlines the role each team member plays throughout the project, and justifies those assignments based on their academic backgrounds, professional experience, and the specific requirements of building an AI-assisted code enforcement system for the Town of Riverdale Park.

1.1 Role Overview

The following table summarizes each team member's assigned role, primary responsibilities, and coordination function within the team. Role assignments reflect each member's academic background, skills, and the functional demands of the project.

Team Member	Role	Responsibilities	Coordination Function
Fechi Iwudyke	Client Liaison & Ethics/Fairness Lead	Manages all written and verbal communication with Ryan Chelton and the Town of Riverdale Park; coordinates meeting agendas and follow-up summaries; leads the fairness and privacy evaluation, including analysis of disparate impact across neighborhoods; ensures deliverables are framed in client-accessible language	Primary point of contact between the team and the client; ensures team outputs align with client expectations and responsible AI requirements; escalates scope or timeline changes to the full team.
Niping (Eileen) Duan	Domain Lead & Evaluation Lead	Applies civil/urban infrastructure expertise to interpret violation code categories and their real-world context; leads dataset exploration and quality assessment; designs and executes the model evaluation framework including per-class metrics and neighborhood-level fairness analysis; validates that	Bridges technical model outputs and domain knowledge; ensures evaluation criteria reflect both machine learning best practices and the operational realities of code enforcement; reviews all technical findings before client delivery.

Team Member	Role	Responsibilities	Coordination Function
		model outputs are meaningful in the civic enforcement context.	
Chihao Li	Technical Lead & ML Engineer	Leads the design, implementation, and iterative improvement of the image classification model; selects model architecture (e.g., ResNet/EfficientNet fine-tuning), manages training pipelines, and oversees preprocessing and data augmentation; draws on prior industry experience as an AI engineer to ensure the solution is technically sound and practically deployable.	Primary technical decision-maker; coordinates with the Evaluation Lead on model performance targets; advises on feasibility constraints and infrastructure requirements; responsible for code quality, reproducibility, and technical documentation.
Jake Sheehi	Documentation Lead	Tracks project timeline, task completion, and team coordination across all phases; prepares and maintains all written deliverables including the project report and responsible AI documentation; leads preparation of the client-facing presentation; ensures all sections meet rubric requirements and submission deadlines.	Keeps the team on schedule and accountable; consolidates contributions from all members into coherent deliverables; serves as a secondary client communication channel and manages logistics for client meetings and presentations.

1.2 Justification of Role Assignments

- Niping (Eileen) Duan is a second-year PhD student in Civil and Environmental Engineering with a research focus on urban infrastructure. Her domain expertise makes her particularly suited to understand the civic and regulatory context of code enforcement, lending credibility and depth to the dataset analysis and evaluation tasks she will lead.
- Jake Sheehi is a fourth-year student at UMD majoring in Information Science. Throughout his time at UMD, he has gained a lot of experience within both documentation and leading a team itself throughout its progress. His responsibilities will include tracking timelines, consolidating deliverables, acting as a secondary channel for client communication, as well as preparing the client presentation when the time comes.
- Chihao Li is a first-year master's student in the Human-Computer Interaction program at the University of Maryland. He previously worked in AI product roles at ByteDance (TikTok) and later as the technical co-founder of an AI startup, where he led the development of large language model fine-tuning and image recognition systems. This background in both AI product design and hands-on model development makes him well-suited to serve as Technical Lead & ML Engineer.

- Fechi Iwudyke is a fourth year undergraduate student at The University of Maryland-College Park studying Information Science with a minor in General Business. My focuses are data analytics in sectors such as business strategy, healthcare, and consumer management. These skills and practices allow me to carry out my role and manage communication pathways from the Town of Riverdale Park to Ryan Chelton. I can also analyze community impact and client accessibility with a fair and ethical standpoint.

These four roles collectively ensure that all major project functions are covered: technical execution, domain grounding, client communication, and quality oversight.

2. Task Identification

The tasks outlined in this section represent a comprehensive, logically sequenced plan for delivering a functional AI-assisted code enforcement prototype to the Town of Riverdale Park. Tasks are scoped to be achievable within a single semester while setting the foundation for the longer-term expansion Ryan Chelton described (3-4 additional municipalities over the following year). The following table presents each task with its estimated timeline, responsible owner(s), and a detailed description of the work involved.

Task	Timeline	Owner	Description
Task #1: Client & Domain Understanding	Weeks 1–2	All Members	Review the full meeting transcript with Ryan Chelton. Study Riverdale Park's town code structure (600–700 violation types). Map which violation types are visually distinguishable from photographs versus those requiring contextual judgment. Identify the 20–30 most frequent violation categories to prioritize model development.
Task #2: Data Access & Secure Transfer	Weeks 2–3	Client Liaison + Tech Lead	Coordinate with Ryan Chelton and the Town's IT department to establish a secure data-sharing protocol. Confirm whether address-level metadata can be shared for fairness analysis. Ensure compliance with FOIA and applicable privacy obligations. Document the data governance agreement in writing.
Task #3: Dataset Exploration & Preprocessing	Weeks 3–4	ML Lead + Data Analyst	Conduct exploratory data analysis (EDA): inspect class distribution, image dimensions, quality variance, and metadata completeness. Apply preprocessing steps including resizing, normalization, and augmentation. Create train/validation/test splits ensuring representative class coverage. Flag low-quality or ambiguous images for client review.
Task #4: Baseline Model Development	Weeks 4–6	ML Lead + Tech Lead	Implement a baseline image classification pipeline using a pre-trained convolutional neural network (e.g., ResNet-50, EfficientNet) fine-tuned on the Riverdale Park dataset. Evaluate on initial held-out test set. Document hyperparameters, architecture choices, and training procedures for reproducibility.
Task #5: Model Evaluation & Fairness Analysis	Weeks 6–8	Evaluation Lead + All Members	Compute standard classification metrics: accuracy, precision, recall, F1-score per violation category. Where address data is available, produce disaggregated performance metrics by neighborhood zone. Identify categories with high false positive rates that could lead to disproportionate enforcement. Present findings to the full team before client delivery.

Task	Timeline	Owner	Description
Task #6: Iterative Model Improvement	Weeks 7–9	ML Lead + Tech Lead	Based on evaluation results, iterate on model architecture, training data, or preprocessing. Explore approaches for handling multi-label images. Consider confidence thresholding: only surface recommendations above a specified confidence level to inspectors. Re-evaluate after each iteration.
Task #7: Prototype System Design	Weeks 8–10	Tech Lead + Client Liaison	Design a lightweight user-facing interface or API endpoint that accepts an uploaded image and returns top-N suspected violation codes with confidence scores. Prototype does not need to be a full production system. A functional demonstration with documentation is acceptable. Align interface design with inspector workflow described by Ryan Chelton.
Task #8: Documentation & Responsible AI Report	Weeks 9–11	All Members	Prepare a technical write-up covering: dataset characteristics, model architecture, evaluation results, fairness findings, limitations, and deployment considerations. Include a responsible AI section addressing potential harms, human oversight design, data privacy protections, and recommendations for future improvement.
Task #9: Client Presentation Preparation	Weeks 11–12	All Members (Client Liaison leads)	Develop a professional slide deck tailored to a non-technical client audience. Translate technical findings into plain language. Prepare to demonstrate the prototype. Rehearse Q&A. Ensure the presentation addresses feasibility, risks, and next steps for operational deployment.
Task #10: Final Delivery & Handoff	Week 12	All Members	Deliver the final written report, prototype or documented system, and presentation to Ryan Chelton and the client. Provide all code and documentation in a format that can be handed off to future teams or the Town's IT department. Confirm whether the client wishes to retain access to any models or data pipelines.

2.1 Task Sequencing and Dependencies

The tasks above follow a logical progression from problem understanding to solution delivery. Tasks T1 and T2 are foundational: without domain knowledge and secure data access, no meaningful technical work can begin. T3 and T4 form the core data-to-model pipeline, while T5 and T6 constitute an iterative evaluation loop. T7 through T10 represent the system integration, documentation, and delivery phase.

Key dependencies include: T3 depends on T2 (data access); T4 depends on T3 (preprocessed data); T5 and T6 depend on T4 (trained model); T7 depends on T5-T6 (validated model); T8-T9 depend on all preceding tasks. This dependency chain means delays in data access (T2) represent the highest scheduling risk for the project.

3. Risk Identification

Developing an AI system for a civic enforcement context involves a range of risks that extend beyond standard machine learning challenges. The risks below were identified through analysis of the client meeting transcript, the project description, and established responsible AI frameworks. For each risk, we describe the potential impact and propose a mitigation strategy.

Risk	Severity	Description & Potential Impact	Mitigation Strategy
Limited Dataset Size	High	With only ~1,000 labeled image-code pairs and 600-700 possible violation codes, many categories will have insufficient training examples. This imbalance can lead to poor model generalization, especially for rare violations.	Prioritize the most common violation categories for the initial model. Use data augmentation to artificially expand underrepresented classes. Explore transfer learning from pre-trained vision models to compensate for limited data.
Image Quality Inconsistency	Medium	As confirmed by Ryan Chelton, images were captured under varying lighting, weather, and compositional conditions. This variability can degrade model accuracy and make it difficult to establish reliable visual features.	Apply image preprocessing pipelines (normalization, contrast adjustment). During evaluation, assess model performance stratified by image condition. Flag low-confidence predictions for mandatory human review.
Multi-Label Ambiguity	Medium	A single photograph may contain evidence of multiple simultaneous violations (e.g., overgrown grass and a vehicle parked on an unimproved surface). A single-label classification approach may miss or misattribute violations.	Design the output layer to support multi-label classification. Clearly document labeling protocol with the client to determine whether images should be annotated with all applicable codes or just the primary one.
Data Access and Privacy	High	Some inspection photos are taken inside private residences as part of rental licensing. Transferring this data outside Town Hall raises legal and ethical concerns under privacy law and the Freedom of Information Act.	Work with Ryan Chelton and the Town's IT department to establish a secure data access arrangement (e.g., on-site access or a university-managed secure server). Never store personal or residential interior images on personal devices or unsecured platforms.
Fairness and Disparate Impact	High	An AI enforcement tool, if not carefully evaluated, could systematically flag properties in lower-income or minority neighborhoods at higher rates, perpetuating historical inequities in code enforcement.	Report disaggregated performance metrics by neighborhood or geographic zone (where address data is available). Adopt a human-in-the-loop design where AI provides recommendations only. Explicitly evaluate false positive rates across different community areas.
Scope Creep / Future Scalability	Low	Ryan indicated plans to expand to 3-4 additional municipalities within a year. Each municipality uses different code standards, potentially requiring model retraining or architecture changes.	Design the system with modularity and scalability in mind from the start. Document the model pipeline so that future teams can adapt it. Clearly scope the current semester deliverable as a Riverdale Park prototype only.
Over-reliance on AI	Medium	If inspectors habitually accept AI suggestions without independent	Communicate clearly in documentation and training materials

Risk	Severity	Description & Potential Impact	Mitigation Strategy
Recommendation s		judgment, the system's errors become institutionalized, undermining the intent of human oversight.	that the AI output is advisory only. Design the interface to require an explicit inspector confirmation step, discouraging passive acceptance.

Of the risks identified above, data access and privacy (Risk 4) and fairness and disparate impact (Risk 5) are considered the most consequential from an ethical and organizational standpoint. These are also the risks that, if not addressed early, could undermine trust in the system and create legal liability for the Town. The team is committed to addressing these proactively through design choices rather than post-hoc corrections.

4. Task Development: Proposed Solution

4.1 Problem Statement

The Town of Riverdale Park's code enforcement staff currently conduct manual inspections, photographing potential violations and manually cross-referencing images against a code library containing 600 to 700 distinct standards. This process is time-consuming and error-prone, particularly for rare or complex violations that inspectors encounter infrequently. Ryan Chelton identified this as the primary bottleneck: staff who see common violations daily can identify them efficiently, but obscure violations require time-intensive code lookup and are occasionally missed entirely.

The proposed solution addresses this bottleneck by developing an AI-powered image classification system that analyzes inspection photographs and returns a ranked list of suspected violation codes, along with the relevant code chapter and section reference. The system is designed as a decision-support tool, not an autonomous enforcement agent. All final judgments remain with the human inspector.

4.2 Proposed Solution

We propose the development of a multi-label image classification system built on transfer learning from a pre-trained convolutional neural network (CNN). The system will:

- Accept a photograph as input from an inspector's mobile device or desktop interface
- Process the image through a fine-tuned vision model trained on the Riverdale Park labeled dataset
- Return the top-N most likely violation codes (e.g., top 3), ranked by confidence score, along with the applicable code chapter and section
- Present the output in a clear, inspector-readable format requiring confirmation or rejection of each suggestion
- Log inspector feedback (agree/disagree) to support future model retraining and continuous improvement

This design reflects Ryan Chelton's stated vision of a "human-in-the-loop" system where AI serves as an assistant rather than a decision-maker. The logging of inspector feedback also addresses his intuition that staff confirmations and rejections could help educate future recommendations.

4.3 Technical Approach

The technical pipeline consists of four main components:

1) Data Preprocessing.

The labeled image-code pairs will undergo quality assessment, duplicate removal, and class distribution analysis. Images will be resized to a standard input dimension, normalized, and augmented using techniques such as random cropping, horizontal flipping, and color jitter to compensate for the limited dataset size and inconsistent capture conditions described by the client.

2) Model Architecture.

We will fine-tune a pre-trained ResNet-50 or EfficientNet-B0 model, architectures that have demonstrated strong performance on small fine-tuning datasets, on the Riverdale Park violation categories. Given the large number of potential violation codes (600-700), we will initially focus on the 20-30 most frequently occurring categories, as these are most likely to have sufficient labeled examples and deliver the most immediate operational value to the client. The output layer will be modified to support multi-label classification, enabling a single image to generate predictions across multiple violation codes simultaneously.

3) Evaluation Framework.

Model performance will be measured using per-class precision, recall, and F1-score, supplemented by top-3 accuracy (whether the correct code appears among the top 3 predictions). We will additionally compute calibration metrics to assess whether the model's confidence scores are reliable. Where address metadata is available and permissible to share, we will stratify performance metrics by neighborhood to evaluate for potential disparate impact.

4) Prototype Interface.

A lightweight prototype will be developed as either a web-based interface or a RESTful API endpoint. The interface will accept an image upload and return the model's top-N predictions with confidence scores and code references. The prototype will require explicit inspector confirmation for each suggestion, and will log all interactions for future training data collection. The prototype is intended as a proof-of-concept rather than a production-grade system.

4.4 Feasibility Assessment

1) Data Availability.

The client has confirmed the existence of approximately 1,000 labeled image-code pairs. While this is a modest dataset for training a computer vision model from scratch, it is sufficient for fine-tuning a pre-trained model. Ryan Chelton also confirmed that he designed the data collection software himself, meaning that additional data collection is feasible if required, and that data can be formatted to meet the team's specifications.

2) Technical Resources.

The team includes a member with professional AI engineering experience at a large technology company, providing the technical foundation necessary for model development. University computing resources (cluster access) are available as a backup if on-site data access cannot be arranged. The primary open-source tools to be used (PyTorch or TensorFlow, scikit-learn, standard CV libraries) are freely available and well-documented.

3) Organizational Capacity.

Ryan Chelton expressed openness to a working prototype as the ideal deliverable, while noting that a thoroughly documented system would also be valuable. This flexible expectation means the team can deliver meaningful value even if technical constraints limit the sophistication of the prototype. The client's willingness to consult with the Town's IT department on secure data access further demonstrates organizational readiness.

4) Timeline Realism.

The 12-week task plan presented in Section 2 allocates approximately two weeks to each major phase (understanding, data, modeling, evaluation, prototyping, documentation/delivery). This timeline is tight but feasible given the team's combined skill set and the bounded scope of the prototype. The largest uncertainty is the data access timeline (T2), which depends on the Town's IT department, a factor outside the team's control.

5) Constraints and Limitations.

The primary constraints are: (1) the small dataset size limits model accuracy for rare violation categories; (2) the inconsistent image capture conditions reduce the reliability of visual features; (3) data privacy requirements may restrict the team's ability to work with the full dataset remotely; and (4) the prototype will be limited to the most common violation categories and will not cover the full 600-700 code library within the semester timeframe.

4.5 Ethical and Responsible AI Considerations

Because this system is intended for use in a regulatory and civic enforcement context, where AI recommendations could influence government actions affecting residents' lives and property, responsible AI design is not optional but central to the project's value. The team will address ethical considerations through the following commitments:

- Human oversight by design: The system will never autonomously issue violations. All AI output will be framed explicitly as a recommendation requiring human confirmation.
- Fairness evaluation: To the extent that address-level metadata is available, the team will evaluate whether the model's error rates are systematically higher in any particular neighborhood or demographic area. This will be reported transparently to the client.
- Data minimization: The team will request only the data necessary for model development and will not retain personal or residential data beyond the project's conclusion.
- Transparency in limitations: The final report and client presentation will clearly communicate the model's known failure modes, confidence thresholds, and categories where human judgment should be given extra weight.
- Documentation for accountability: All model training decisions, evaluation results, and data handling protocols will be documented so that the Town can audit the system and make informed deployment decisions.

Ryan Chelton noted during the client meeting that he believes a human-in-the-loop design substantially mitigates fairness concerns. The team agrees that this architectural choice is necessary, but notes that it is not sufficient on its own: the team will also produce a fairness audit as part of the evaluation phase (T5) to ensure the recommendation system does not embed systematic bias.

4.6 Implementation Next Steps

Following submission of this report, the immediate next steps are:

1. Confirm secure data access arrangement with Ryan Chelton and the Town's IT department (T2).

2. Receive and inventory the initial dataset of labeled image-code pairs.
3. Complete domain analysis: identify the 20-30 highest-priority violation categories and confirm labeling conventions with the client (T1).
4. Begin exploratory data analysis and preprocessing pipeline development (T3).
5. Reconvene with Ryan Chelton on April 8th to report initial findings and confirm alignment on deliverable scope.

5. Client Presentation Plan

5.1 Presentation Structure

The team will prepare a professional, client-facing slide deck for presentation to Ryan Chelton and any other Town of Riverdale Park stakeholders. The presentation will be structured to be accessible to a non-technical audience, avoiding jargon and grounding all explanations in concrete, real-world examples from the Riverdale Park context.

The proposed slide structure is as follows:

- Slide 1 — Title and Team Introduction: Project name, team members, and client name.
- Slide 2 — Problem Statement: What is the current challenge? Why does it matter for the Town's inspectors and residents?
- Slide 3 — Proposed Solution Overview: Plain-language description of the AI recommendation system. A simple process diagram showing the inspector workflow with AI integration.
- Slide 4 — What the Data Looks Like: Summary of the dataset. Representative examples of the types of violations the system will learn to identify (with any privacy-sensitive content removed).
- Slide 5 — Task Plan & Timeline: Visual timeline (Gantt-style) showing each task phase and its expected duration. Key milestones highlighted.
- Slide 6 — Risk Overview: 3–4 key risks in plain language, with the mitigation approaches we are taking.
- Slide 7 — Fairness and Privacy Commitments: How the team is approaching responsible AI. Why human oversight is built into the design.
- Slide 8 — What We Will Deliver: Working prototype demonstration (if ready), documented system, evaluation report with fairness metrics.
- Slide 9 — Next Steps and Questions: Immediate actions needed from the client (data access, IT coordination). Open Q&A.

5.2 Communication Approach

The presentation will be delivered in a conversational tone appropriate for a government client audience. Technical terms will be introduced with brief, accessible explanations. Visuals will be used liberally to replace text wherever possible.

The team will practice the full presentation at least once before the client meeting on April 8th, with each team member responsible for presenting the sections most closely related to their role. Fechi Iwudyke, as Client Liaison, will open and close the presentation and manage the Q&A. The team will prepare a one-page handout summarizing key deliverables and next steps that Ryan Chelton can retain after the meeting.

Following the presentation, the team will circulate a brief written summary of any new client feedback or revised requirements within 48 hours, ensuring that all team members and the supervising professor are aligned on any changes to scope or approach.